

Data Science: Capstone Lightning Talk #1

Pitch & Problem Statements

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1. Turbofan Engine Pressure Prediction

Problem & Goals

- Physical aviation sensors degrade or fail under extreme conditions.
- **Goal:** Predict HPC Outlet Static Pressure using operational settings and active sensors.
- **Approach:** Build a fail-safe "virtual sensor" via regression/Random Forest/ XGBoost models.

Data Overview

- **Source:** NASA C-MAPSS Turbofan Engine Data.
- **Scale:** 20,630 rows, 26 structured columns.
- **Basic EDA:** rewrote column names, checked dtypes, isnulls, and target distribution



Column	Type	Description
engine	int	Engine No.
cycle	int	Time, In Cycles
sensor8	float	Physical Fan Speed (rpm)
sensor9	float	Physical Core Speed (rpm)
sensor10	float	Engine Pressure Ratio (P50/P2)
sensor11	float	target: HPC Outlet Static Pressure (psia)

2. Steam Game Success Prediction

Problem & Goals

- > Most Steam games fail to gain traction, while a select few go viral.
- > **Goal:** Forecast Peak Concurrent Users (CCU) based on release metadata and features.
- > **Approach:** Handle right-skewed data with log-transforms and apply Random Forest/XGBoost.

Data Overview

- > **Source:** Kaggle Steam Games Dataset.
- > **Scale:** 125,855 rows, 39 columns.
- > **Basic EDA:** reassigned column names due to right side shift, checked dtypes, isnulls, and target distribution



Column	Type	Description
AppID	int	Unique Steam application ID
Name	object	Game title
Release date	object	Release date on Steam
Estimated owners	object	Estimated owner range (e.g., 0 - 20000)
Peak CCU	int	Target: Peak concurrent users
Required age	float	Minimum required age to play

3. Aviation Accident Severity

Problem: Identifying higher-risk flight conditions before they turn fatal is vital for safety.

Goal & Approach: Classify accident severity from structured NTSB incident data using Random Forest.

Data: NTSB Aviation Accident Database (1962-2023). Contains 88,889 rows and 31 columns.

Basic EDA: checked `value_counts`, `dtypes`, `isnulls`, and target distribution



Column	Type	Description
Event.Id	object	Unique identifier for the NTSB event
Investigation.Type	object	Indicates if the event was an Accident or an Incident
Airport.Code	object	3- or 4-letter code of the associated airport (if applicable)
Airport.Name	object	Name of the associated airport (if applicable)
Injury.Severity	object	Target: Highest level of injury severity (e.g., Fatal, Serious, Minor)

4. Weather Temperature Prediction

Problem & Goals

Weather directly impacts aviation and mechanical performance. Unpredictable conditions introduce systemic risks.

Predict apparent temperature from atmospheric measurements like humidity, wind, pressure, and visibility.

Approach & Data

Clean anomalous zero-values, run Linear Regression baselines, and develop robust Random Forest predictors.

Data: Weather in Szeged, Hungary (2006-2016).

Contains 96.5k hourly observations and 12 columns.

Basic EDA: renamed Precip type column, checked dtypes, isnulls, and value_counts



Column	Type	Description
Formatted Date	object	Timestamp of the hourly observation
Summary	object	Short weather summary category
Precip Type	object	rain / snow
Temperature (C)	float	air temperature in Celsius
Apparent Temperature (C)	float	target: 'Feels like' temperature

Summary of Topics

1. Engine Pressure

Building a fail-safe virtual sensor model to predict HPC Outlet Static Pressure using simulated C-MAPSS data.

2. Steam Game CCU

Analyzing release metadata, tags, and game features to predict viral peak concurrent player counts.

3. Aviation Accidents

Classifying crash injury severity using NTSB structured operational and weather incident data.

4. Weather Prediction

Predicting apparent temperature outcomes by leveraging humidity, wind speed, pressure, and visibility.